

REFERRING PHYSICIAN

DUCHARME, NICOLA L ND
928 FT. STOCKTON DR #213
SAN DIEGO, CA 92103

MUNGUIA, HANNAH
DOB: 08/18/1995
Gender: FEMALE
Accession: A997177
Patient ID: 261790

Collected: 03/13/2024 UNK
Received: 03/20/2024 13:48
Reported: 04/09/2024 14:34
Reprinted: 04/10/2024 13:55
Amended:
Corrected:

TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
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BORRELIOSIS - Lyme Disease

Lyme ImmunoBlot IgM Serum
IGX Criteria: Negative
CDC/NYS Criteria: Negative

[REVISED REPORT: EFFECTIVE APRIL 10, 2019]

Lyme ImmunoBlot IgM detects antibodies to B. burgdorferi strains and species

Band (kDa)	23*	31*	34*	39*	41*	93
Intensity	-	-	-	-	-	-

Band Intensity: Positive: + to +++++, Indeterminate: Ind, Negative: (-)

INTERPRETATION

IGX CRITERIA

CDC/NYS CRITERIA

Positive

2 or more of the starred bands are present (+): 23*, 31*, 34*, 39*, 41* kDa

2 or more of the following bands are present (+): 23*, 39*, 41* kDa

Negative

Does not meet IGX criteria for a positive.

Does not meet CDC/NYS criteria for a positive.

LIMITATION: Bands 31* and 34* kDa are present in Lyme vaccinated patients. Viral antibodies cross react with the 93 kDa antigen.

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LYME IMMUNOBLOT IGM SPECIATION

Revised report:06-09-22

B. burgdorferi US(genus) Serum Negative
B. burgdorferi US Spp Serum Negative

*BbSs: Borrelia burgdorferi sensu stricto includes but not limited to
B. burgdorferi B31 and B. burgdorferi 297.
BbSl: Borrelia burgdorferi sensu lato includes but not limited to B.
mayonii and B. californiensis.*

B. burgdorferi European(genus) Serum Negative
B. burgdorferi European Spp Serum Negative

*Borrelia burgdorferi European species include but not limited to B.
afzelii, B. garinii, B. spielmanii and B. valaisiana.*

Testing performed at IGeneX 556 Gibraltar Drive Milpitas CA 95035 (800) 832-3200

Diagnosis should not be based on laboratory results alone. Results should be interpreted in conjunction with clinical symptoms and patient history.

NOTE: Western Blots, ImmunoBlots, Lyme Dot Blot, Epitope, PCR, IFA, FISH, C. pneumoniae IgG/IgA, CD57, IGXSpot, Broad Coverage Antibody, COVID-19 Test - These tests were developed and their performance characteristics determined by IGeneX, Inc. They have not been cleared or approved by the FDA. The FDA has determined that such approval is not necessary. These tests are used for clinical purposes and should not be regarded as investigational or for research. IGeneX, Inc. is licensed by CMS and NYS to perform high complexity clinical laboratory testing.

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Lyme ImmunoBlot IgG Serum
IGX Criteria: Negative
CDC/NYS Criteria: Negative

[REVISED REPORT: EFFECTIVE APRIL 10, 2019]

Lyme ImmunoBlot IgG detects antibodies to B. burgdorferi strains and species

Band (kDa)	18	23*	28	30	31*	34*	39*	41*	45	58	66	93*
Intensity	-	-	-	-	-	-	-	+	-	-	-	-

Band Intensity: Positive: + to +++++, Indeterminate: Ind, Negative: (-)

INTERPRETATION	IGX CRITERIA	CDC/NYS CRITERIA
Positive	2 or more of the starred bands are present (+): 23*, 31*, 34*, 39*, 41*, 93* kDa	5 or more of the following bands are present (+): 18, 23*, 28, 30, 39*, 41*, 45, 58, 66, 93* kDa
Negative	Does not meet IGX criteria for a positive.	Does not meet CDC/NYS criteria for a positive.

LIMITATION: Bands 31* and 34* kDa are present in Lyme vaccinated patients. Viral antibodies cross react with the 93 kDa antigen.

LYME IMMUNOBLOT IGG SPECIATION

Revised report:06-09-22

B. burgdorferi US(genus) Serum Negative
B. burgdorferi US spp Serum Negative

BbSs: *Borrelia burgdorferi sensu stricto* includes but not limited to

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B. burgdorferi B31 and *B. burgdorferi* 297.

BbSI: *Borrelia burgdorferi* sensu lato includes but not limited to *B.*

mayonii and *B. californiensis*.

B. burgdorferi European(genus) Serum Negative

B. burgdorferi European Spp Serum Negative

Borrelia burgdorferi European species include but not limited to *B.*

afzelii, *B. garinii*, *B. spielmanii* and *B. valaisiana*.

LYME SCREEN IMMUNOASSAY IGM

Lyme Screen IgM Serum Negative Negative

LYME SCREEN IMMUNOASSAY IGG

Lyme Screen IgG Serum Negative Negative

BORRELIOSIS- Relapsing Fever Borrelia

TBRF Borrelia ImmunoBlot IgM

TBRF Borrelia genus ImmunoBlot IgM Serum IND

Positive: Detected 2 or more TBRF Borrelia species-specific antibodies.
Indeterminate: Detected only 1 TBRF Borrelia species-specific antibody.
Negative: No TBRF Borrelia specific antibody detected.

* TBRF ImmunoBlot IgM detects antibodies to *Borrelia hermsii*, *B.*

turicatae, *B. miyamotoi* and TBRF Borrelia species.

TBRF Borrelia ImmunoBlot IgM Species

B. miyamotoi Serum Negative

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TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
B. hermsii	Serum	Negative		
B. turicatae	Serum	Negative		
TBRF Borrelia spp	Serum	IND		

TBRF Borrelia ImmunoBlot IgG

TBRF Borrelia genus ImmunoBlot IgG Serum	Negative	Positive: Detected 2 or more TBRF Borrelia species-specific antibodies. Indeterminate: Detected only 1 TBRF Borrelia species-specific antibody. Negative: No TBRF Borrelia specific antibody detected.
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* TBRF ImmunoBlot IgG detects antibodies to Borrelia hermsii, B. turicatae, B. miyamotoi, B. coriaceae and TBRF Borrelia species.

TBRF Borrelia ImmunoBlot IgG Species

B. miyamotoi	Serum	Negative
B. hermsii	Serum	Negative
B. turicatae	Serum	Negative
TBRF Borrelia spp	Serum	Negative

BABESIOSIS

BABESIA IMMUNOBLOT IGM

Babesia ImmunoBlot IgM(genus)	Serum	Negative	Positive: Presence of 2 or more Babesia specific antibodies. Negative: Presence of only 1 or no Babesia specific antibody.
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* Babesia ImmunoBlot test detects specific IgM antibodies to Babesia species, including but not limited to B. microti, B. duncani and B. divergens.

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TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
B. microti ImmunoBlot IgM	Serum	Negative		
B. duncani ImmunoBlot IgM	Serum	Negative		
Babesia spp.	Serum	Negative		

Limitations:

Negative test result does not exclude possibility of Babesia infection, may retest in 6-8 weeks.

Results should be interpreted in conjunction with clinical symptoms and other laboratory findings.

BABESIA IMMUNOBLOT IGG

Babesia ImmunoBlot IgG(genus)

Serum

Positive

Positive: Presence of 2 or more Babesia specific antibodies.

Negative: Presence of only 1 or no Babesia specific antibody.

** Babesia ImmunoBlot test detects specific IgG antibodies to Babesia species, including but not limited to B. microti, B. duncani and B. divergens.*

B. microti ImmunoBlot IgG

Serum

Negative

B. duncani ImmunoBlot IgG

Serum

Positive

Babesia spp.

Serum

Negative

Limitations:

Negative test result does not exclude possibility of Babesia infection, may retest in 6-8 weeks.

Results should be interpreted in conjunction with clinical symptoms and other laboratory findings.

EHRLICHIOSIS

HME IFA - IgM

Serum

<20

< 20 : Negative

= 20 : May or may not indicate active infection

Titer

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TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
>=40 : Indicates active infection				
HME IFA - IgG	Serum	<40	< 40 : Negative < 160 : May or may not suggest active infection >=160 : Indicates active infection	Titer
<u>ANAPLASMOSIS</u>				
HGA IFA - IgM	Serum	40	< 20 : Negative = 20 : May or may not indicate active infection >=40 : Indicates active infection	Titer
HGA IFA - IgG	Serum	<40	< 40 : Negative < 160 : May or may not suggest active infection >=160 : Indicates active infection	Titer
<u>BARTONELLOSIS</u>				
BARTONELLA IMMUNOBLOT IGM Bartonella genus	Serum	IND	Positive: Detected 2 or more Bartonella genus or species-specific antibody. Indeterminate: Detected only 1 Bartonella genus or species- specific antibody. Negative: No Bartonella specific antibody detected.	
B. elizabethae	Serum	Negative		
B. vinsonii	Serum	Negative		

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TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
B. henselae	Serum	Negative		
B. quintana	Serum	Negative		
Bartonella species	Serum	IND		

Report revision: Effective 11-15-2022

Bartonella ImmunoBlot test detects specific antibodies to

Bartonella species, including but not limited to B.elizabethae, B.

vinsonii, B.henselae, B.quintana, B.bacilliformis, B.clarridgeiae, B.

grahamii, B.koehlerae, B.washoensis and B.rochalimae.

BARTONELLA IMMUNOBLOT IGG

Bartonella genus	Serum	Positive	Positive: Detected 2 or more Bartonella genus or species-specific antibody. Indeterminate: Detected only 1 Bartonella genus or species-specific antibody. Negative: No Bartonella specific antibody detected.
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B. elizabethae	Serum	Negative
B. vinsonii	Serum	Positive
B. henselae	Serum	Negative
B. quintana	Serum	Negative
Bartonella species	Serum	Negative

Report revision: Effective 11-15-2022

Bartonella ImmunoBlot test detects specific antibodies to

Bartonella species, including but not limited to B.elizabethae, B.

vinsonii, B.henselae, B.quintana, B.bacilliformis, B.clarridgeiae, B.

grahamii, B.koehlerae, B.washoensis and B.rochalimae.

RICKETTSIOSIS

R. rickettsii IFA - IgG	Serum	<40	< 40 : Negative < 160 : May or may not suggest active infection ≥160 : Indicates active infection	Titer
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TEST	SPECIMEN	RESULT	REFERENCE RANGE	UNITS
R. typhi IFA - IgG	Serum	<40	< 40 : Negative < 160 : May or may not suggest active infection ≥160 : Indicates active infection	Titer



End of Report

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